Lecture 3 — Green's functions

defined w.r.t $\boxed{\mathcal{H} = H - \mu N}$!

1. Definitions

$\mathcal{U}(t,t')$ → time evolution operator

$$\hat{O}_{\mathcal{H}}(t) = \mathcal{U}^{\dagger}(t,t_0)\,\hat{O}\,\mathcal{U}(t,t_0) \rightarrow \text{Heisenberg represent. of } \hat{O} \qquad (1)$$

~~$\psi_{\mathcal{H}}(1) = \psi(x_1 t_1)$, $x = (\vec{r}\sigma)$ → field operators; $\psi^{\dagger}_{\mathcal{H}}(1) = \psi^{\dagger}(x_1 t_1)$~~

~~$\psi_{\mathcal{H}}$ $O_{H}(t)$~~ If $\hat{O}$ depends on x, $\hat{O}_{\mathcal{H}} \rightarrow \hat{O}_{\mathcal{H}}(x,t) \equiv \hat{O}_{\mathcal{H}}(1)$ — for convenience

$\hat{T}_C$ – orders in chronological order (time ↑ from R → L), but changes sign if two ψ or $\psi^{\dagger}$ permuted to honor times.

- Time-ordered GF ("causal")

$$G(1,2) = -\frac{i}{\hbar}\,\frac{\mathrm{Tr}\left(\mathcal{U}(t_0 - i\beta\hbar, t_0)\,\hat{T}_C[\psi_{\mathcal{H}}(1)\,\psi^{\dagger}_{\mathcal{H}}(2)]\right)}{\mathrm{Tr}\left(\mathcal{U}(t_0 - i\beta\hbar, t_0)\right)} \equiv -\frac{i}{\hbar}\langle T_C[\psi_{\mathcal{H}}(1)\psi^{\dagger}_{\mathcal{H}}(2)]\rangle \qquad (2)$$

$\hbar = 1$ in the following.

$$\boxed{G(1,2) = -i\,\langle T_C[\psi_{\mathcal{H}}(1)\,\psi^{\dagger}_{\mathcal{H}}(2)]\rangle} \qquad (3)$$

since $\psi^{\dagger}_H$ and ψ_H were permuted!

$$= \theta(t_1 - t_2)\underbrace{(-i)\langle\psi_{\mathcal{H}}(1)\,\psi^{\dagger}_{\mathcal{H}}(2)\rangle}_{\boxed{G^{>}(1,2)}\ \text{"greater" GF}} + \theta(t_2 - t_1)\underbrace{(+i)\langle\psi^{\dagger}_{\mathcal{H}}(2)\,\psi_{\mathcal{H}}(1)\rangle}_{\boxed{G^{<}(1,2)}\ \text{"lesser" GF}} \qquad (4)$$

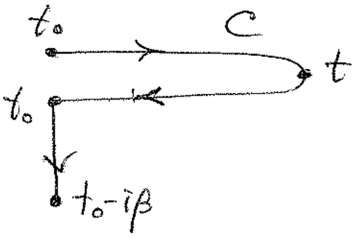
- Retarded/Advanced

$$G^{R}(1,2) = -i\,\theta(t_1 - t_2)\,\langle\{\psi_{\mathcal{H}}(1), \psi^{\dagger}_{\mathcal{H}}(2)\}\rangle \qquad (5)$$

$$G^{A}(1,2) = +i\,\theta(t_2 - t_1)\,\langle\{\psi_{\mathcal{H}}(1), \psi^{\dagger}_{\mathcal{H}}(2)\}\rangle \qquad (6)$$

$$\left.\begin{aligned} G^{R} &= (G^{>} - G^{<})\,\theta(t_1 - t_2) \\ G^{A} &= (-G^{>} + G^{<})\,\theta(t_2 - t_1) \end{aligned}\right\} \rightarrow \boxed{G^{R} - G^{A} = G^{>} - G^{<}} \qquad (7)$$

- Others



$G(1,2)$ $\rightarrow$ anywhere on the contour C

1 ∈ real, 2 ∈ imaginary → ~~G(1,2)~~ 2 is "later" than 1, →

$$G(1,2) \equiv i\langle \psi_{\mathcal{H}}^{\dagger}(2)\psi_{\mathcal{H}}(1)\rangle \equiv G^{<}(1,2) = G^{<}(t_1, \underbrace{t_0 - i\tau}_{t_2}) = G^{\rceil}(t_1,\tau_2) \equiv G^{\rceil}(1,2) \quad (8)$$

2 ∈ real, 1 ∈ imaginary →

$$G(1,2) \equiv -i\langle \psi_{\mathcal{H}}(1)\psi_{\mathcal{H}}(2)\rangle =: G^{>}(1,2) \equiv G^{>}(t_0 - i\tau, t_2) = G^{\lceil}(1,2) \quad (9)$$

1,2 ∈ imaginary →

Matsubara GF

$$G(1,2) = G(t_0 - i\tau_1, t_0 - i\tau_2) = \theta(\tau_1-\tau_2)\,G^{>}(t_0-i\tau_1, t_0-i\tau_2) + \theta(\tau_2-\tau_1)\,G^{<}(t_0-i\tau_1, t_0-i\tau_2) \equiv G^{M}(1,2) \quad (10)$$

2. Boundary conditions ($x_{1,2}$ are omitted for convenience of notations)

$$G^{>}(t_0-i\beta, t_2) \sim \mathrm{Tr}\left[\mathcal{U}(t_0-i\beta,t_0)\,\psi_{\mathcal{H}}(t_0-i\beta)\,\psi_{\mathcal{H}}^{\dagger}(2)\right]$$

$$= \mathrm{Tr}\Big[\underbrace{\mathcal{U}(t_0-i\beta,t_0)\,\mathcal{U}(t_0,t_0-i\beta)}_{1}\,\psi(x_1)\,\mathcal{U}(t_0-i\beta,t_2)\,\psi^{\dagger}(x_2)\,\mathcal{U}(t_2,t_0)\Big]$$

$$= \mathrm{Tr}\Big[\underbrace{\mathcal{U}(t_0-i\beta,t_2)}_{\mathcal{U}(t_0-i\beta,t_0)\,\mathcal{U}(t_0,t_2)}\,\psi^{\dagger}(x_2)\,\mathcal{U}(t_2,t_0)\,\psi(x_1)\Big] = \mathrm{Tr}\left[\mathcal{U}(t_0-i\beta,t_0)\,\psi_{\mathcal{H}}^{\dagger}(2)\,\psi_{\mathcal{H}}(x_1,t_0)\right]$$

$\hookrightarrow -G^{<}(x_1,t_0;2)$, i.e. $\boxed{G^{>}(t_0-i\beta,t_2) = -G^{<}(t_0,t_2)}$, (11)

valid for any t_2. ~~If t_2~~ Since $t_0 - i\beta$ is the "largest" time possible, then $G(t_0-i\beta,t_2) \equiv G^{>}(t_0-i\beta,t_2)$. Since t_0 is "smallest" time possible, then $G(t_0,t_2) = +G^{<}(t_0,t_2)$, ⇒ (11) is valid for the whole G:

$$\boxed{G(t_0-i\beta,2) = -G(t_0,2)} \quad (12)$$

Similarly, $\boxed{G(1,t_0) = -G(1,t_0-i\beta)}$ (13)

- Many-particle GF: $G(1\ldots n;1'\ldots n') = \left(\frac{1}{i\hbar}\right)^{n}\langle T_C[\psi_{\mathcal{H}}(1)\ldots\psi_{\mathcal{H}}(n)\,\psi_{\mathcal{H}}^{\dagger}(1')\ldots\psi_{\mathcal{H}}^{\dagger}(n')]\rangle$ (14)

3. Some useful properties

$$G^>(1,2)^* = \left[-i \left\langle \psi_{\mathcal{H}}(1)\, \psi^\dagger_{\mathcal{H}}(2) \right\rangle\right]^* \sim +i\,\mathrm{Tr}\left[e^{\beta \mathcal{H}_0}\, \psi_{\mathcal{H}}(1)\psi^\dagger_{\mathcal{H}}(2)\right]^*$$

$$= i\,\mathrm{Tr}\left[\psi_{\mathcal{H}}(2)\, \psi^\dagger_{\mathcal{H}}(1)\, e^{-\beta \mathcal{H}_0}\right] = i\,\mathrm{Tr}\left[e^{-\beta \mathcal{H}_0}\, \psi_{\mathcal{H}}(2)\, \psi^\dagger_{\mathcal{H}}(1)\right]$$

$$\rightarrow -G^>(2,1) \quad, \text{or} \qquad G^>(1,2)^* = -G^>(2,1)$$

Here we used that

$$\left(\psi_{\mathcal{H}}(1)\right)^\dagger = \left(U(t_0,t_1)\,\psi(x_1)\,U(t_1,t_0)\right)^\dagger =$$

$$= U^\dagger(t_1,t_0)\,\psi^\dagger(x_1)\,U^\dagger(t_0,t_1) = U(t_0,t_1)\,\psi^\dagger(x_1)\,U(t_1,t_0) \equiv \psi^\dagger_{\mathcal{H}}(1)$$

and similarly for $\left(\psi^\dagger_{\mathcal{H}}(2)\right)^\dagger = \psi_{\mathcal{H}}(2)$.

This kind of property is valid for both $G^<$ and $G^>$:

$$\boxed{G^{\gtrless}(1,2)^* = -G^{\gtrless}(2,1)} \tag{15}$$

This means that $G^>(1,2)$ may only need to be calculated for $t_1 > t_2$; for $t_1 < t_2$ it follows from (15). Similarly for $G^<$.

• ~~Similarly:~~

$$\cancel{G^\urcorner(t,\tau) = G^<(t, t_0-\beta\tau)}, \qquad \cancel{G^\ulcorner(\tau,t) = G^>(t_0-\beta\tau, t)}$$

$$\hookrightarrow \cancel{G^\urcorner(t,\tau)^* = G^<(t,t_0-\beta\tau)^* = -G^<(t_0-\beta\tau,t) = G^\ulcorner}$$

• We have 9 functions:

$$\begin{array}{cc} & \begin{array}{ccc} \rightarrow & \leftarrow & \downarrow \end{array} \\ \begin{array}{c} \rightarrow \\ \leftarrow \\ \downarrow \end{array} & \begin{pmatrix} G^c & G^< & G^\urcorner \\ G^> & G^a & G^\urcorner \\ G^\ulcorner & G^\ulcorner & G^M \end{pmatrix} \end{array} \qquad G^c, G^a \rightarrow \text{consist of } \underbrace{G^>, G^<}_{\text{need only these two}} \tag{16}$$

anticasual

casual

• $G^\ulcorner$ and $G^\urcorner$ are also related:

$$G^\ulcorner(t_0-i\tau, t) = G^>(t_0-i\tau, t) = -i\left\langle \psi_{\mathcal{H}}(x_1, t_0-i\tau)\, \psi^\dagger_{\mathcal{H}}(x_2,t)\right\rangle$$

$$\sim -i\,\mathrm{Tr}\Big[e^{-\beta \mathcal{H}_0} \underbrace{U(t_0, t_0 - i\tau)}_{\exp[+i\mathcal{H}_0((t_0 - i\tau) - t_0)] = e^{\tau \mathcal{H}_0}} \psi(x_1) \underbrace{U(t_0 - i\tau, t_0)}_{e^{-\tau \mathcal{H}_0}} \psi^\dagger_{\mathcal{H}}(x_2, t) \Big]$$

(since on the imaginary axis $\mathcal{H} \to \mathcal{H}_0$!)

$$= -i\,\mathrm{Tr}\Big[e^{\beta \mathcal{H}_0} e^{\tau \mathcal{H}_0} \psi(x_1) e^{-\tau \mathcal{H}_0} \psi^\dagger_{\mathcal{H}}(x_2 t) \Big]$$

$$\hookrightarrow\ G^{\lceil}(t_0 - i\tau, t)^* = +i\,\mathrm{Tr}\Big[\psi_{\mathcal{H}}(x_2 t)\, e^{-\tau \mathcal{H}_0} \psi^\dagger(x_1) e^{\tau \mathcal{H}_0} e^{\beta \mathcal{H}_0} \Big] \qquad (17)$$

Consider now

$$G^{\rceil}(t, t_0 - i\tau) = +i \langle \psi^\dagger_{\mathcal{H}}(x_1, t_0 - i\tau)\, \psi_{\mathcal{H}}(x_2, t) \rangle$$

$$\sim i\,\mathrm{Tr}\Big[e^{-\beta \mathcal{H}_0} e^{\tau \mathcal{H}_0} \psi^\dagger(x_1) e^{-\tau \mathcal{H}_0} \psi_{\mathcal{H}}(x_2 t) \Big]$$

$$= i\,\mathrm{Tr}\Big[\psi_{\mathcal{H}}(x_2 t)\, e^{-(\beta - \tau)\mathcal{H}_0} \psi^\dagger(x_1)\, e^{(\beta-\tau)\mathcal{H}_0} e^{-\beta \mathcal{H}_0} \Big]$$

It is the same as (17) for $\tau \to \beta - \tau$:

$$\boxed{G^{\lceil}(t_0 - i\tau, t)^* = G^{\rceil}(t, t_0 - i(\beta - \tau))} \qquad (18)$$

4. Matsubara Gf $\to$ on imaginary axis ($\mathcal{H} \to \mathcal{H}_0$)

- $G_M(x_1, t_0 - i\tau_1; x_2, t_0 - i\tau_2) \equiv G_M(\tau_1, \tau_2)$ $\qquad \boxed{0 < \tau_1, \tau_2 < \beta}$

$$= \theta(\tau_1 - \tau_2)\, G^>(\tau_1, \tau_2) + \theta(\tau_2 - \tau_1)\, G^<(\tau_1, \tau_2)$$

where

$$G^>(\tau_1, \tau_2) = G^>(x_1, t_0 - i\tau_1; x_2, t_0 - i\tau_2) = -i \langle \psi_{\mathcal{H}_0}(x_1, t_0 - i\tau_1)\, \psi^\dagger_{\mathcal{H}_0}(x_2, t_0 - i\tau_2) \rangle$$

$$\approx -i\,\mathrm{Tr}\Big\{ e^{-\beta \mathcal{H}_0} \underbrace{U(t_0, t_0 - i\tau_1)}_{e^{\tau_1 \mathcal{H}_0}} \psi(x_1) \underbrace{U(t_0 - i\tau_1, t_0)}_{e^{-\tau_1 \mathcal{H}_0}} \underbrace{U(t_0, t_0 - i\tau_2)}_{e^{\tau_2 \mathcal{H}_0}} \psi^\dagger(x_2) \underbrace{U(t_0 - i\tau_2, t_0)}_{e^{-\tau_2 \mathcal{H}_0}} \Big\}$$

$$= -i\,\mathrm{Tr}\Big\{ e^{-\beta \mathcal{H}_0} e^{\tau_1 \mathcal{H}_0} \psi(x_1)\, e^{-(\tau_1 - \tau_2)\mathcal{H}_0} \psi^\dagger(x_2)\, e^{-\tau_2 \mathcal{H}_0} \Big\}$$

and depends only on $\tau_1-\tau_2$:

$$G^>(\tau_1,\tau_2) = -i\,\frac{\mathrm{Tr}\left[e^{-\beta\mathcal{H}_0}\, e^{(\tau_1-\tau_2)\mathcal{H}_0}\,\psi(x_1)\, e^{-(\tau_1-\tau_2)\mathcal{H}_0}\,\psi^\dagger(x_2)\right]}{\mathrm{Tr}\left(e^{-\beta\mathcal{H}_0}\right)} \qquad (19)$$

similarly,

$$G^<(\tau_1,\tau_2) = i\,\frac{\mathrm{Tr}\left[e^{-\beta\mathcal{H}_0}\, e^{(\tau_2-\tau_1)\mathcal{H}_0}\,\psi^\dagger(x_2)\, e^{-(\tau_2-\tau_1)\mathcal{H}_0}\,\psi(x_1)\right]}{\mathrm{Tr}\left(e^{-\beta\mathcal{H}_0}\right)} \qquad (20)$$

also depends on $(\tau_1-\tau_2)$ $\rightarrow$ $G_M(\tau_1,\tau_2)$ only depends on "times" difference.
↳ Fourier integral (ω-representation)

- Lehmann representation $\boxed{Z = \mathrm{Tr}\left(e^{-\beta\mathcal{H}_0}\right)}$, $\boxed{\mathcal{H}_0|\lambda\rangle = E_\lambda|\lambda\rangle}$ (21)

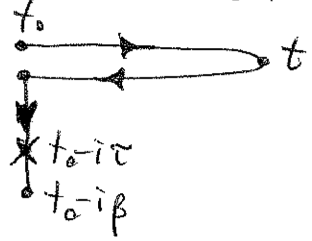
$$G^>(\tau_1,\tau_2) = \frac{-i}{Z}\sum_{\lambda\lambda'} e^{-\beta E_\lambda}\, e^{(\tau_1-\tau_2)E_\lambda}\,\langle\lambda|\psi(x_1)|\lambda'\rangle\langle\lambda'|\psi^\dagger(x_2)|\lambda\rangle\, e^{-(\tau_1-\tau_2)E_{\lambda'}}$$

$$= -\frac{i}{Z}\sum_{\lambda\lambda'} e^{-\beta E_\lambda}\, e^{(\tau_1-\tau_2)(E_\lambda-E_{\lambda'})}\,\langle\lambda|\psi(x_1)|\lambda'\rangle\langle\lambda'|\psi^\dagger(x_2)|\lambda\rangle \qquad (22)$$

and

$$G^<(\tau_1,\tau_2) = \frac{i}{Z}\sum_{\lambda\lambda'} e^{-\beta E_\lambda}\, e^{(\tau_2-\tau_1)(E_\lambda-E_{\lambda'})}\,\langle\lambda|\psi^\dagger(x_2)|\lambda'\rangle\langle\lambda'|\psi(x_1)|\lambda\rangle \qquad (23)$$

- Antiperiodicity condition



$$G^<_M(x_1 t_0; x_2 t_0 - i\tau) = \frac{i}{Z}\sum_{\lambda\lambda'} e^{-\beta E_\lambda}\, e^{+\tau(E_\lambda-E_{\lambda'})}\,\psi^\dagger_{\lambda\lambda'}(x_2)\,\psi_{\lambda'\lambda}(x_1) \quad \boxed{0<\tau<\beta} \qquad (24)$$

Now consider

$$G^>(x_1 t_0 - i\beta; x_2 t_0 - i\tau) = -\frac{i}{Z}\sum_{\lambda\lambda'} e^{-\beta E_\lambda}\, e^{(\beta-\tau)(E_\lambda-E_{\lambda'})}\,\psi_{\lambda\lambda'}(x_1)\,\psi^\dagger_{\lambda'\lambda}(x_2) = |\lambda\leftrightarrow\lambda'|$$

$$= -\frac{i}{Z}\sum_{\lambda\lambda'} e^{-\beta E_{\lambda'}}\, e^{(\beta-\tau)(E_{\lambda'}-E_\lambda)}\,\psi_{\lambda'\lambda}(x_1)\,\psi^\dagger_{\lambda\lambda'}(x_2)$$

$$= -\frac{i}{Z}\sum_{\lambda\lambda'} e^{-\beta E_\lambda}\, e^{-\tau(E_{\lambda'}-E_\lambda)}\,\psi^\dagger_{\lambda\lambda'}(x_2)\,\psi_{\lambda'\lambda}(x_1) \equiv -G^<(x_1 t_0; x_2 t_0 - i\tau)$$

$$\boxed{G^>(t_0 - i\beta, t_0 - i\tau) \equiv -G^<(t_0; t_0 - i\tau)} \qquad (25)$$

Similarly,

$$\boxed{G^{>}(t_0-i\tau, t_0-i\beta) = -G^{<}(t_0-i\tau, t_0)} \qquad (26)$$

$\hookrightarrow$ $G_M(t_0, t_0-i\tau) = G^{<}(t_0, t_0-i\tau) = -G^{>}(t_0-i\beta, t_0-i\tau)$

$G_M(t_0-i\beta, t_0-i\tau) \equiv G^{>}(t_0-i\beta, t_0-i\tau)$

$$\hookrightarrow \boxed{G_M(t_0, t_0-i\tau) = -G_M(t_0-i\beta, t_0-i\tau)} \leftarrow \text{antiperiodicity} \qquad (27)$$

Also:

$$\boxed{G_M(t_0-i\tau, t_0) = -G_M(t_0-i\tau, t_0-i\beta)} \quad \text{for } \underline{\underline{\forall\, 0<\tau<\beta}} \qquad (28)$$

- G_M depends on $\tau = \tau_1 - \tau_2$. Since $\tau_1, \tau_2 \in [0, \beta]$, then $\tau \in [-\beta, \beta]$.

$\tau < 0 \to \tau_1 < \tau_2 \to G_M(\tau_1 \tau_2) = G_M(\tau<0) \equiv G^{<}(\tau<0)$

$\equiv G^{<}(t_0, t_0+i\tau) \overset{(25)}{\equiv} -G^{>}(t_0-i\beta, t_0+i\tau) \equiv -G_M(\beta, -\tau)$

$$= -G_M(\beta+\tau) \to \boxed{G_M(\tau<0) = -G_M(\underbrace{\tau+\beta}_{})} \qquad (29)$$

$0 < \tau+\beta < \beta$

$\hookrightarrow$ positive

$\hookrightarrow$ $\boxed{G_M(\tau) \text{ is periodic with the period of } 2\beta.}$

$\boxed{-\beta < \tau < \beta} \to$ Fourier series:

$$\boxed{G_M(\tau) = \sum_{z_\nu} e^{-i\tau z_\nu} G_M(iz_\nu)}, \quad \boxed{z_\nu = \frac{\pi\nu}{\beta}}, \quad \nu = 0, \pm1, \pm2, \ldots \qquad (30)$$

$$\boxed{G_M(iz_\nu) = \frac{1}{2\beta}\int_{-\beta}^{\beta} e^{iz_\nu\tau} G_M(\tau)\, d\tau} \qquad (31)$$

To satisfy (29):

$$\sum_{z_\nu} e^{-i\tau z_\nu} G_M(iz_\nu) \equiv -\sum_{z_\nu} e^{-i(\tau+\beta)z_\nu} G_M(iz_\nu) = \sum_{z_\nu} e^{-i\tau z_\nu} G_M(iz_\nu)\left(-e^{-i\beta z_\nu}\right)$$

$$\hookrightarrow -e^{-i\beta z_\nu} = 1 \to e^{-i\pi\nu} = -1 \to \boxed{\nu = \text{odd}} \qquad (32)$$

5. Tight-binding Hamiltonian

$$\mathcal{H} = \sum_{nm} (h_{nm} - \mu N_{nm})\, a_n^\dagger a_m = \sum_{nm} \mathcal{E}_{nm} a_n^\dagger a_m \qquad (33)$$

$\mathcal{E} = \| \mathcal{E}_{nm} \| \to$ diagonalised : $\mathcal{E}_{nm} = \langle n | \hat{h} - \mu \hat{N} | m \rangle$ (ϕ_n, ϕ_m)

Choose another basis set:

$$\chi_i = \sum_n \underbrace{U_{in}}_{\text{unitary}} \phi_n \Rightarrow \phi_n = \sum_i U_{in}^{*} \chi_i$$

$\hookrightarrow$ $\mathcal{E}_{nm} = \sum_{ij} U_{in}^{*} U_{jm} \langle i | \hat{h} - \mu \hat{N} | j \rangle$

and
$$\mathcal{H} = \sum_{ij} \underbrace{\left(\sum_n U_{in}^{*} a_n^\dagger\right)}_{c_i^\dagger} \underbrace{\left(\sum_m U_{jm} a_m\right)}_{c_j} \underbrace{\langle i | \hat{h} - \mu \hat{N} | j \rangle}$$

new operators

Choose the new basis χ_i to be eigenvectors of $\hat{h} - \mu \hat{N}$

$\hookrightarrow$ $\boxed{\mathcal{H} = \sum_i \bar{\mathcal{E}}_i c_i^\dagger c_i}$, $\bar{\mathcal{E}}_i$ – eigenvectors of $\hat{h} - \mu \hat{N}$. $\qquad (34)$
U – eigenvectors standing by columns.

So, it is sufficient to consider (34) rather than (33). $\boxed{\text{free particles}}$

6. $G^{>}$ for the ~~TB Hami~~ free particles

$$G^{>}(x_1 t_1 ; x_2 t_2) = -i \langle \psi(x_1 t_1) \psi^\dagger(x_2 t_2) \rangle = -i \sum_{jj'} \chi_j(x_1) \chi_{j'}^{*}(x_2) \times$$
$$\times \langle c_j(t_1) c_{j'}^\dagger(t_2) \rangle . \qquad (35)$$

$$c_j(t) = e^{i\mathcal{H}(t-t_0)} c_j e^{-i\mathcal{H}(t-t_0)} \qquad (\mathcal{H} \text{ – does } \underline{\text{not}} \text{ depend on } t) \quad (36)$$

$$\frac{d}{dt} c_j(t) = e^{i\mathcal{H}(t-t_0)} [i\mathcal{H} c_j - i c_j \mathcal{H}] e^{-i\mathcal{H}(t-t_0)} = i e^{i\mathcal{H}(t-t_0)} [\mathcal{H}, c_j] e^{-i\mathcal{H}(t-t_0)}$$

$$[\mathcal{H}, c_j] = \sum_i \bar{\mathcal{E}}_i (c_i^\dagger c_i c_j - \underbrace{c_j c_i^\dagger c_i}_{(\delta_{ij} - c_i^\dagger c_j) c_i}) = \sum_i \bar{\mathcal{E}}_i (c_i^\dagger c_i c_j - \delta_{ij} c_i + c_i^\dagger \underbrace{c_j c_i}_{-c_i c_j})$$

$$= -\bar{\mathcal{E}}_j c_j \to \frac{d}{dt} c_j(t) = -i \bar{\mathcal{E}}_j c_j(t)$$

Integrating $\rightarrow$ $c_j(t) = f_j \, e^{-i\bar{\varepsilon}_j t}$

$\hookrightarrow$ $c_j(0) = c_j \Rightarrow f_j = c_j$ $\quad \rightarrow \quad \boxed{c_j(t) = c_j \, e^{-i\bar{\varepsilon}_j t}}$ (27)

Similarly, $\boxed{c_j^\dagger(t) = c_j^\dagger \, e^{+i\bar{\varepsilon}_j t}}$ (28)

$$\hookrightarrow \langle c_j(t_1)\, c^\dagger_{j'}(t_2)\rangle = e^{-i\bar{\varepsilon}_j t_1} e^{i\bar{\varepsilon}_{j'} t_2} \langle c_j c_j^\dagger \rangle$$

- Consider

$$Z = \mathrm{Tr}\left(e^{-\beta \mathcal{H}}\right) = \sum_\lambda \langle \lambda | e^{-\beta\mathcal{H}} | \lambda \rangle$$

$$|\lambda\rangle = |\{n_i\}\rangle = |0101100\ldots\rangle$$

$$\mathcal{H}|\lambda\rangle = E_\lambda |\lambda\rangle, \quad E_\lambda = \sum_i n_i \bar{\varepsilon}_i, \quad \bar{\varepsilon}_i = \varepsilon_i - \mu$$

$$\hookrightarrow Z = \sum_{n_1=0}^{1} \sum_{n_2=0}^{1} \cdots \langle \{n_i\}| e^{-\beta\mathcal{H}} |\{n_i\}\rangle = \sum_{n_1}\sum_{n_2}\cdots e^{-\beta \sum_k \bar{\varepsilon}_k n_k}$$

$$= \prod_i \left(\sum_{n=0}^{1} e^{-\beta \bar{\varepsilon}_i n} \right) = \prod_i \left(1 + e^{-\beta\bar{\varepsilon}_i}\right) \quad \leftarrow \text{the partition function} \quad (29)$$

- Consider now

$$\langle c_j^\dagger c_j \rangle = \frac{1}{Z} \mathrm{Tr}\left(e^{-\beta\mathcal{H}} \overbrace{c_j^\dagger c_j}^{\hat{n}_j} \right) = \frac{1}{Z} \sum_{n_1=0}^{1}\sum_{n_2=0}^{1}\cdots e^{-\beta\sum_k \bar{\varepsilon}_k n_k} \langle \{n_i\} | \hat{n}_j | \{n_i\} \rangle$$

$$= \frac{1}{Z} \sum_{n_1=0}^{1}\sum_{n_2=0}^{1}\cdots e^{-\beta\sum_k \bar{\varepsilon}_k n_k}\, n_j = \frac{1}{Z}\left[\prod_{i\neq j} \left(\sum_{n=0}^{1} e^{-\beta\bar{\varepsilon}_i n} \right) \right] \sum_{n_j=0}^{1} n_j e^{-\beta \bar{\varepsilon}_j n_j}$$

$$= \frac{e^{-\beta\bar{\varepsilon}_j}}{e^{-\beta\bar{\varepsilon}_j} + 1} = \left[e^{\beta\bar{\varepsilon}_j} + 1 \right]^{-1} \leftarrow \text{Fermi function} \quad (30)$$

$$\hookrightarrow \langle c_j^\dagger c_j \rangle = f(\bar{\varepsilon}_j), \quad \langle c_j c_j^\dagger \rangle = \langle 1 - c_j^\dagger c_j \rangle = 1 - f(\bar{\varepsilon}_j) \quad (31)$$

- Hence,

$$G^>(1,2) = -i \sum_{jj'} \chi_j(x_1)\chi^*_{j'}(x_2)\,\delta_{jj'} \left(1 - \overbrace{f(\bar{\varepsilon}_j)}^{f_j}\right) \times e^{-i\bar{\varepsilon}_j t_1} e^{i\bar{\varepsilon}_j t_2} \cancel{= -i\sum_j \chi_j(x)\chi_j^*(x)(1-f_j)}$$

$$\boxed{G^>(1,2) = -i \sum_j \chi_j(x_1)\chi_j^*(x_2)(1-f_j)\, e^{-i\bar{\varepsilon}_j(t_1 - t_2)}} \quad (32)$$

- Similarly,

$$\boxed{G^<(1,2) = i \sum_j \chi_j(x_1)\chi_j^*(x_2) f_j e^{-i\bar{\varepsilon}_j(t_1-t_2)}} \qquad (33)$$

7. Retarded / Advanced G for "free" electrons.

$$G^R(1,2) = \theta(t_1-t_2)(G^> - G^<) = \theta(t_1-t_2)\sum_j \chi_j(x_1)\chi_j^*(x_2) e^{-i\bar{\varepsilon}_j(t_1-t_2)} \times$$

$$\times\left[-i(1-f_j) - i f_j\right] = -i\theta(t_1-t_2)\sum_j \chi_j(x_1)\chi_j^*(x_2) e^{-i\bar{\varepsilon}_j(t_1-t_2)} \qquad (34)$$

$$G^A(1,2) = i\theta(t_2-t_1)\sum_j \chi_j(x_1)\chi_j^*(x_2) e^{-i\bar{\varepsilon}_j(t_1-t_2)} \qquad (35)$$

8. Matsubara GF for "free" electrons

$$G_M^>(\tau_1,\tau_2) = -i\langle \psi_{H_0}(t_0-i\tau_1)\psi_{H_0}^+(t_0-i\tau_2)\rangle,$$

$$\psi_{H_0}(t_0-i\tau_1) = U(t_0,t_0-i\tau_1)\psi U(t_0-i\tau_1,t_0) \equiv e^{\tau_1 H_0}\psi e^{-\tau_1 H_0}$$

$\to$ take $G^>$ on the real part of the contour and replace $it_1 \to \tau_1$, $it_2 \to \tau_2$:

$$G_M^>(\tau_1,\tau_2) = -i\sum_j \chi_j(x_1)\chi_j^*(x_2)(1-f_j) e^{-\bar{\varepsilon}_j(\tau_1-\tau_2)}, \quad \tau_1>\tau_2 \qquad (36)$$

$$G_M^<(\tau_1,\tau_2) = i\sum_j \chi_j(x_1)\chi_j^*(x_2) f_j e^{-\bar{\varepsilon}_j(\tau_1-\tau_2)}, \quad \tau_1<\tau_2 \qquad (37)$$

9. Other GF's

$$G^\lceil(x_1 t_0-i\tau_1; x_2 t_2) = -i\langle \psi_{H_0}(t_0-i\tau_1)\psi_{H\to H_0 \text{ in this case}}^+(t_2)\rangle$$

where only τ_1 replaces it_1 of $G^>$ on the real contours:

$$G^\lceil(\tau_1,t_2) = -i\sum_j \chi_j(x_1)\chi_j^*(x_2)(1-f_j) e^{-\bar{\varepsilon}_j(\tau_1-it_2)} \qquad (38)$$

$$G^\rceil(t_1,t_0-i\tau_2) = G^\rceil(t_1,\tau_2) = i\sum_j \chi_j(x_1)\chi_j^*(x_2) f_j e^{-\bar{\varepsilon}_j(it_1-\tau_2)} \qquad (39)$$

So, in the TB case any GF can be written explicitly via difference of times. Hence, their FT can be considered, for instance,

$$G^<(x_1,x_2;\omega) = \int \frac{d(t_1-t_2)}{2\pi}\, G^<(x_1t_1;x_2t_2)\, e^{i\omega(t_1-t_2)}$$

$$= i\sum_j \chi_j(x_1)\,\chi_j^*(x_2)\, f_j\, \delta(\omega-\bar{\varepsilon}_j)\,, \quad \bar{\varepsilon}_j = \varepsilon_j - \mu \qquad (40)$$

↳ the form of a DOS. Similarly for others.

8. The form of GF convenient for the perturbation theory.

- Interaction representation

$$\boxed{\mathcal{H} = \mathcal{H}_0' + V + H_t'} \qquad (41)$$

$\mathcal{H}_0'$: TB, contains μ

V → two-particle interaction, no time dependence

H_t' → all t-dependence is here (eg. external field)

At time $t=0$, $H_{t=0}'=0$, so $\boxed{\mathcal{H}_0 = \mathcal{H}_0' + V}$ ← initial Hamiltonian $(t=0)$

$\boxed{W = V + H_t'}$ → perturbation

- ~~We introduce~~ Need to rewrite $u(t_0-i\beta, t_0)$. Hence, introduce:

$$\boxed{\hat{v}(t) = e^{it\mathcal{H}_0'}\, e^{-it\mathcal{H}_0}} \qquad \hat{v}(t)\,\hat{v}^\dagger(t) = 1 \qquad (42)$$

Also $\boxed{\hat{v}(t,t') = \hat{v}(t)\,\hat{v}^\dagger(t')} = e^{it\mathcal{H}_0'}\, e^{-it\mathcal{H}_0}\, e^{it'\mathcal{H}_0}\, e^{-it'\mathcal{H}_0'} \qquad (43)$

$$i\partial_t \hat{v}(t,t') = i e^{it\mathcal{H}_0'} \underbrace{(i\mathcal{H}_0' - i\mathcal{H}_0)}_{-iV} e^{-it\mathcal{H}_0}\, e^{it'\mathcal{H}_0}\, e^{-it'\mathcal{H}_0'}$$

$$= \underbrace{\left(e^{it\mathcal{H}_0'}\, V\, e^{-it\mathcal{H}_0'}\right)}_{\tilde{V}_{\mathcal{H}_0'}(t)\ \to\ \text{Heisenberg representation}} \underbrace{e^{-it\mathcal{H}_0'}\, e^{it\mathcal{H}_0'}}_{\text{since } = 1} \underbrace{e^{-it\mathcal{H}_0}\, e^{it'\mathcal{H}_0}\, e^{-it'\mathcal{H}_0'}}_{\hat{v}(t,t')\ \text{again}}$$

↳ $$\boxed{i\partial_t \hat{v}(t,t') = \tilde{V}_{\mathcal{H}_0'}(t)\, \hat{v}(t,t')} \qquad (44)$$

Similarly, $$\boxed{-i\partial_{t'} \hat{v}(t,t') = \hat{v}(t,t')\, \tilde{V}_{\mathcal{H}_0'}(t')} \qquad (45)$$

and their solutions are:

$$\boxed{\begin{aligned}\hat{v}(t,t') &= \hat{T}\exp\left[-i\int_{t'}^{t}\tilde{V}_{\mathcal{H}_0'}(\tau)d\tau\right], \quad t>t'\\ &= \hat{\bar{T}}\exp\left[+i\int_{t}^{t'}\tilde{V}_{\mathcal{H}_0'}(\tau)d\tau\right], \quad t<t'\end{aligned}} \qquad (46)$$

• Also, $\boxed{\hat{v}(t,t')\hat{v}(t',t'') = \hat{v}(t,t'')}$ ← from Def (47)

$\boxed{\hat{v}(t,t) = 1}$ $\boxed{\hat{v}(t,t')^{+} = \hat{v}(t',t)}$ (48)

• Now consider

$$e^{-\beta\mathcal{H}_0} = e^{-\beta\mathcal{H}_0'}\underbrace{\left(e^{\beta\mathcal{H}_0'}e^{-\beta\mathcal{H}_0}\right)}_{\text{can be expressed via } \hat{v}(t,t');}$$ ← which is $u(t_0-i\beta,t_0)$

Consider:

$$v(t_0-i\beta,t_0) = v(t_0-i\beta)v^{+}(t_0)$$

where

$$v(t_0-i\beta) = e^{i(t_0-i\beta)\mathcal{H}_0'}e^{-i(t_0-i\beta)\mathcal{H}_0} = e^{it_0\mathcal{H}_0'}\left(e^{\beta\mathcal{H}_0'}e^{-\beta\mathcal{H}_0}\right)e^{-it_0\mathcal{H}_0}$$

$$v^{+}(t_0) = e^{it_0\mathcal{H}_0'}e^{-it_0\mathcal{H}_0},$$

so that

$$e^{\beta\mathcal{H}_0'}e^{-\beta\mathcal{H}_0} = e^{-it_0\mathcal{H}_0'}v(t_0-i\beta)e^{+it_0\mathcal{H}_0}$$

$$= e^{-it_0\mathcal{H}_0'}v(t_0-i\beta)\underbrace{v^{+}(t_0)v(t_0)}_{1}e^{+it_0\mathcal{H}_0}$$

$$= e^{-it_0\mathcal{H}_0'}\underbrace{v(t_0-i\beta)v^{+}(t_0)}_{v(t_0-i\beta,t_0)}e^{it_0\mathcal{H}_0'} = e^{-it_0\mathcal{H}_0'}v(t_0-i\beta,t_0)e^{it_0\mathcal{H}_0'} \qquad (49)$$

↳

$$e^{-\beta\mathcal{H}_0} = e^{-\beta\mathcal{H}_0'}e^{-it_0\mathcal{H}_0'}v(t_0-i\beta,t_0)e^{it_0\mathcal{H}_0'} \qquad (50)$$

• Introduse

$$\boxed{S(t,t') = e^{it\mathcal{H}_0'}u(t,t')e^{-it'\mathcal{H}_0'}} \qquad (51)$$

~~Then~~ $\boxed{S(t,t')S(t',t'') = S(t,t'')}$, $\boxed{S(t,t) = 1}$ $\boxed{S(t,t')^{+} = S(t',t)}$ (52)

• Then:

$$Tr\left(e^{-\beta\mathcal{H}_0}\right) = Tr\left(e^{-\beta\mathcal{H}_0'}v(t_0-i\beta,t_0)\right)$$ ← denominator of GF (53)

Now the nominator of the GF (assume $t_1 > t_2$):

$$\mathrm{Tr}\left[e^{-\beta \mathcal{H}_0} \mathcal{U}(t_0,t_1)\, \psi(x_1)\, \mathcal{U}(t_1,t_2)\, \psi^\dagger(x_2)\, \mathcal{U}(t_2,t_0) \right]$$

$$= \mathrm{Tr}\Big\{ \left(e^{-\beta \mathcal{H}_0'} e^{-it_0 \mathcal{H}_0'} \mathcal{V}(t_0 - i\beta, t_0)\, e^{it_0 \mathcal{H}_0'} \right) \mathcal{U}(t_0,t_1) \underbrace{\left(e^{-it_1 \mathcal{H}_0'} e^{it_1 \mathcal{H}_0'} \right)}_{\text{since } = 1} \psi_{x_1} \times$$

$$\times \left(e^{-it_1 \mathcal{H}_0'} e^{it_1 \mathcal{H}_0'} \right) \mathcal{U}(t_1,t_2) \left(e^{-it_2 \mathcal{H}_0'} e^{it_2 \mathcal{H}_0'} \right) \psi^\dagger_{x_2} \times$$

$$\times \left(e^{-it_2 \mathcal{H}_0'} e^{it_2 \mathcal{H}_0'} \right) \mathcal{U}(t_2,t_0) \left(e^{-it_0 \mathcal{H}_0'} e^{it_0 \mathcal{H}_0'} \right) \Big\}$$

$$= \mathrm{Tr}\left\{ e^{-\beta \mathcal{H}_0'} \mathcal{V}(t_0 - i\beta, t_0)\, S(t_0,t_1)\, \tilde{\psi}_{\mathcal{H}_0'}(x_1 t_1)\, S(t_1,t_2)\, \tilde{\psi}^\dagger_{\mathcal{H}_0'}(x_2 t_2)\, S(t_2,t_0) \right\} \tag{54}$$

~~Therefore~~, The same for $t_1 < t_2$. Here $\tilde{\psi}_{\mathcal{H}_0'}(1) \to \psi(x_1)$ in the Heisenberg picture with respect to $\mathcal{H}_0'$ only (interaction picture).

- <u>EoM for $S(t,t')$</u>

$$i\partial_t S(t,t') = i e^{it\mathcal{H}_0'} \left(i\mathcal{H}_0' \mathcal{U}(t,t') + \partial_t \mathcal{U}(t,t') \right) e^{-it'\mathcal{H}_0'}$$

$$= e^{it\mathcal{H}_0'} \left(-\mathcal{H}_0' \mathcal{U}(t,t') + \mathcal{H}\, \mathcal{U}(t,t') \right) e^{-it'\mathcal{H}_0'}$$

$$= e^{it\mathcal{H}_0'} \left(V + H_t' \right) \mathcal{U}(t,t')\, e^{-it'\mathcal{H}_0'} \equiv \underbrace{\left(\widetilde{V + H_t'} \right)^t_{\mathcal{H}_0'}}_{\bar{V}_{\mathcal{H}_0'}(t)} S(t,t')$$

$\leftarrow$ Heisenberg picture wrt $\mathcal{H}_0'$ of the rest of $\hat{\mathcal{H}}$, vs $V + H_t'$

$$\hookrightarrow \boxed{i\partial_t S(t,t') = \bar{V}_{\mathcal{H}_0'}(t)\, S(t,t')} \tag{55}$$

Similarly, $$\boxed{-i\partial_{t'} S(t,t') = S(t,t')\, \bar{V}_{\mathcal{H}_0'}(t')} \tag{56}$$

They are solved via $\hat{T}$-exponents:

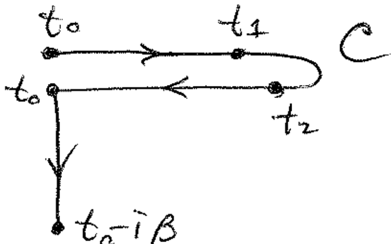
$$\boxed{\begin{aligned} S(t,t') &= \hat{T} \exp\left(-i \int_{t'}^{t} \bar{V}_{\mathcal{H}_0'}(\tau)\, d\tau \right), \quad t > t' \\ &= \tilde{T} \exp\left(+i \int_{t}^{t'} \bar{V}_{\mathcal{H}_0'}(\tau)\, d\tau \right), \quad t < t' \end{aligned}} \tag{57}$$

- Now we are ready to rewrite G; S-operators are combined into a single T-ordered exponent (similarly to the proof for $\hat{O}$ in Lecture 2):

$$\underbrace{\mathcal{U}(t_0-i\beta,t_0)}_{\text{"later" than } t_0} = T\exp\left(-i\int_{t_0}^{t_0-i\beta}\underbrace{\tilde{V}_{\mathcal{H}_0'}(\tau)}_{\text{does not contain } H_t'}d\tau\right) \tag{58}$$

$$\boxed{G(1,2) = -i\,\frac{\mathrm{Tr}\left\{e^{-\beta\mathcal{H}_0'}\,\hat{T}_c\left[\exp\left(-i\int_C \tilde{W}_{\mathcal{H}_0'}(\tau)d\tau\right)\tilde{\psi}_{\mathcal{H}_0'}(x_1t_1)\,\tilde{\psi}^\dagger_{\mathcal{H}_0'}(x_2t_2)\right]\right\}}{\mathrm{Tr}\left\{e^{-\beta\mathcal{H}_0'}\,\hat{T}_c\left[\exp\left(-i\int_C \tilde{W}_{\mathcal{H}_0'}(\tau)d\tau\right)\right]\right\}}} \tag{59}$$

where



and W contains H_t' on the real part of the contour, and does not on the complex one.

This form is convenient for doing perturbation expansion w.r.t. W. Based on Wick's theorem, resulting in diagrams.

9. Relation to observables/useful quantities

- Electron density: $\hat{O} \Rightarrow \hat{\rho}(x)$

$\hat{\rho}(x) = \psi^\dagger(x)\psi(x)$, ~~$\langle\hat{\rho}(x)\rangle_t = \langle\psi^\dagger(x)\psi(x)\rangle$~~

~~$\langle\hat{\rho}(x)\rangle_t = \mathrm{Tr}\left[\mathcal{U}(t_0-i\beta,t)\,\mathcal{U}(t_0,t)\,\psi^\dagger_{\mathcal{H}}(xt)\right)$~~

$$\rho(x) = \langle\hat{\rho}(x)\rangle_t = \langle\psi^\dagger_{\mathcal{H}}(xt)\,\psi_{\mathcal{H}}(xt)\rangle \equiv -i\,G(xt,xt^+) = -iG(1,1^+) \tag{60}$$

$t^+ = t+\delta$ ← "later" than t

- Energy & Grand potential → when we learn eq. of motions for the GF.